Deep Learning Guided Discovery and FEP Validation of NOD2 Binders for the R702W Crohn's Variant

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INTRODUCTION

Background

Over **3 million Americans** suffer from **Crohn's disease**, a painful, incurable inflammatory bowel condition.

The **NOD2 gene** is the strongest genetic risk factor. The **R702W mutation** affects ~**15% of patients**, increasing disease risk **3-fold**. ***No approved drugs directly target the NOD2 protein itself.***

NOD2 is an innate immune sensor that recognizes bacterial muramyl dipeptide (**MDP**) and activates **NF-κB signaling** to mount an antimicrobial response essential for intestinal bacterial clearance and immune homeostasis.

Impaired NOD2 causes chronic intestinal inflammation, the defining hallmark of Crohn's disease.

The R702W mutation lies **79.4 Å** from the binding site, suggesting **allosteric mechanisms** drive its clinical impact. Identifying compounds that maintain binding to R702W could ***restore innate immune function in mutation carriers***.

Research Question

Can computational methods identify compounds that directly bind and modulate NOD2?

No approved drugs target NOD2 directly. Current therapies (biologics, immunosuppressants, steroids) suppress the immune system downstream, treating symptoms rather than the root cause.

This project explores whether **structure-based virtual screening** can identify NOD2-binding compounds that maintain affinity for **both wild-type and R702W mutant** forms of the protein.

Hypothesis: (1) CNN-based docking + ML can prioritize NOD2 binders
(2) MD simulations will confirm stable binding (3) FEP can quantify mutation effects on affinity

Motivation

As an R702W carrier myself, I wanted to understand why this mutation matters for drug response.

The R702W variant is one of the most common NOD2 mutations in Crohn's patients, yet we don't fully understand how it affects potential therapeutics at the molecular level.

This project aims to bridge that gap, moving toward **precision medicine** that accounts for patient genetics.

NOD2 Structure and Pathway

MDP
Muramyl Dipeptide

Smallest immunostimulatory bacterial fragment
Derived from peptidoglycan cell wall
Present in ALL bacteria (Gram+ and Gram−)

Detected intracellularly by NOD2 receptor
Stereoselective recognition (L-D isomer only)

→ Binds NOD2 LRR Domain
→ Triggers NF-κB Activation
→ Initiates Immune Response

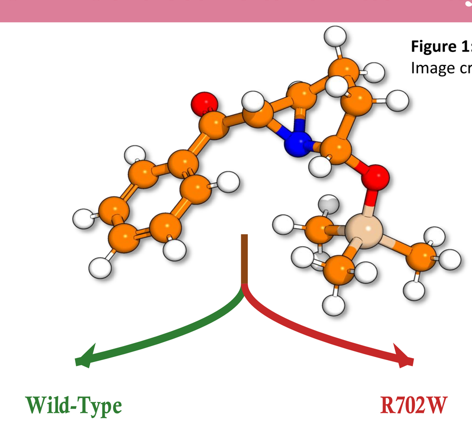
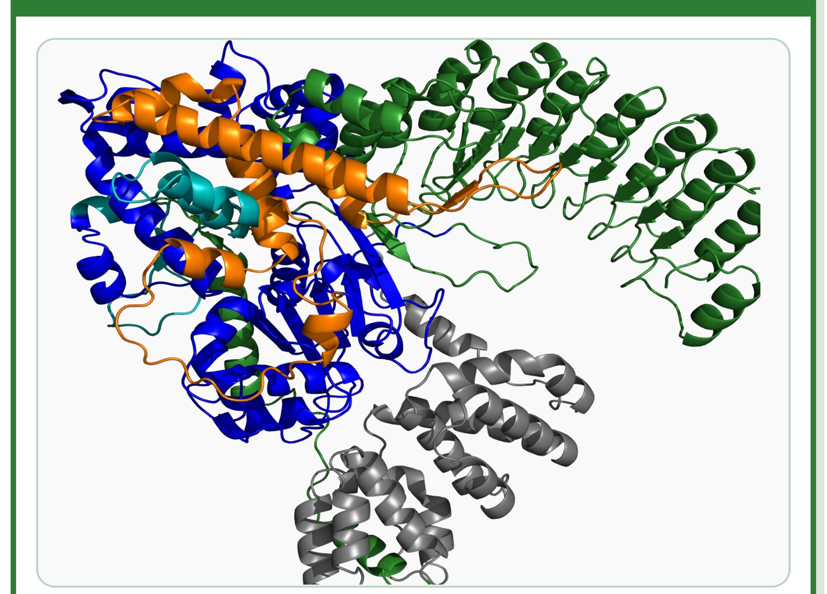


Figure 1: MDP ball-and-stick structure and wild-type/R702W pathway arrow diagram. Image created by finalist using PyMOL and Microsoft PowerPoint, 2026.

Wild-Type NOD2

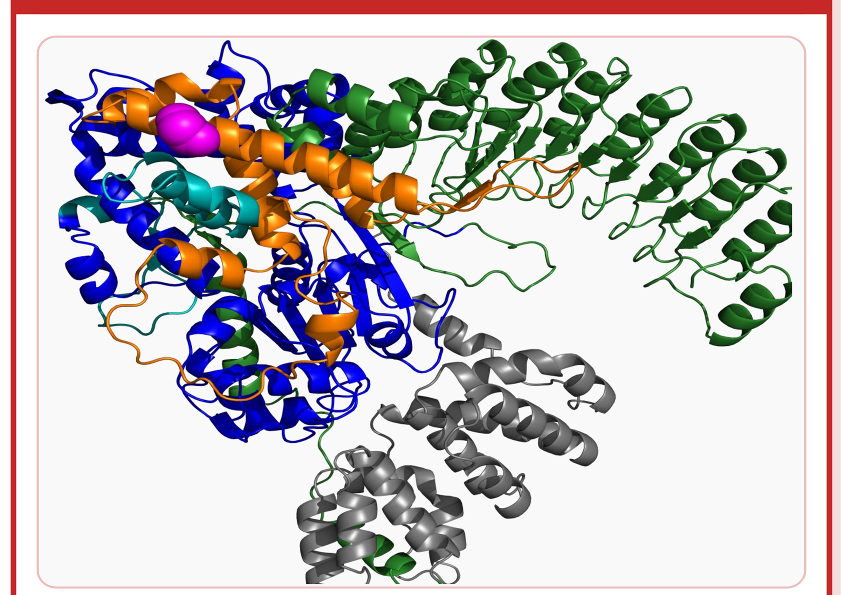


DOMAIN ARCHITECTURE
CARD | NBD | HD1 | HD2 | LRR

MDP binding triggers conformational change
→ CARD domain recruits RIPK2

Figure 2: Wild-type NOD2 full-length protein. Image created by finalist using PyMOL, 2026.

R702W Mutant NOD2

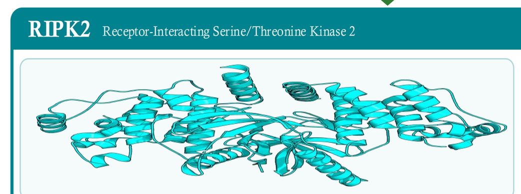


DOMAIN ARCHITECTURE
CARD | NBD | HD1 | HD2 | LRR | Mut

R702W in HD2 allosterically impairs MDP sensing
→ Weakened RIPK2 recruitment (3× Crohn's risk)

Figure 3: R702W mutant NOD2 with mutation site. Image created by finalist using PyMOL, 2026.

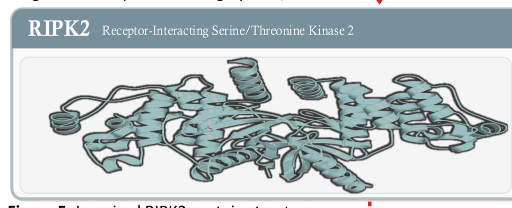
RIPK2 Receptor-Interacting Serine/Threonine Kinase 2 — ✓ ACTIVE



NOD2 CARD domain binds RIPK2 CARD domain
RIPK2 autophosphorylates
→ Recruits TAK1 complex

Figure 4: Active RIPK2 protein structure. Image created by finalist using PyMOL, 2026.

RIPK2 Receptor-Interacting Serine/Threonine Kinase 2 — ⚠ IMPAIRED



Defective NOD2 fails to properly recruit RIPK2
Reduced activation
→ Weakened TAK1 recruitment

Figure 5: Impaired RIPK2 protein structure. Image created by finalist using PyMOL, 2026.

TAK1 TGF-β Activated Kinase 1 — ✓ ACTIVE

RIPK2 recruits TAK1-TAB complex via ubiquitination
TAK1 phosphorylates IKK complex
→ Activates NF-κB pathway

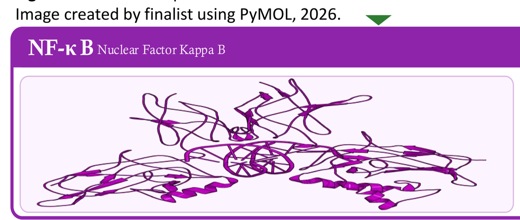
Figure 6: Active TAK1 protein structure. Image created by finalist using PyMOL, 2026.

TAK1 TGF-β Activated Kinase 1 — ⚠ IMPAIRED

Reduced RIPK2 activity fails to fully recruit TAK1 complex
Diminished IKK phosphorylation
→ Weakened NF-κB activation

Figure 7: Impaired TAK1 protein structure. Image created by finalist using PyMOL, 2026.

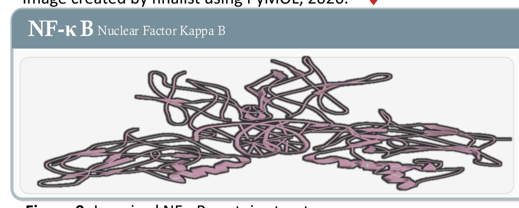
NF-κB Nuclear Factor Kappa B — ✓ ACTIVE



IKK phosphorylates IκB releasing NF-κB to nucleus
Transcribes inflammatory genes
→ TNF-α, IL-1β, IL-6, IL-8

Figure 8: Active NF-κB protein structure. Image created by finalist using PyMOL, 2026.

NF-κB Nuclear Factor Kappa B — ⚠ IMPAIRED



Reduced IKK activation fails to fully release NF-κB
Defective immune gene expression
→ Impaired bacterial clearance

Figure 9: Impaired NF-κB protein structure. Image created by finalist using PyMOL, 2026.

✓ **EFFECTIVE IMMUNE RESPONSE**
Bacterial clearance • Cytokine production (TNF-α, IL-1β, IL-6) • Intestinal homeostasis maintained

X **CROHN'S DISEASE**
Defective bacterial clearance • Chronic inflammation • 3× increased disease risk • Intestinal dysbiosis

METHODS

AlphaFold — NOD2 LRR
e-Drug3D — 2,152 FDA drugs
COCONUT — 7,414 natural products

INCREASING RIGOR →

GNINA — 9,566 compounds
CNN-scored molecular docking on LRR domain. Docking alone has ~0.5 correlation with experiment: ML reranking learns 3D pose features classical scoring functions miss

NOD2-SCOUT — 2,129 compounds
XGBoost ML reranks hits (AUC 0.85-0.93). A strong binder is useless if it can't reach the target or causes organ toxicity: next step applies drug-likeness filters

ADMET — 148 compounds
Lipinski Ro5, Veber, PAINS, and Brenk safety filters. Computation can't capture everything: human review of docking pose geometry, scaffold diversity, and synthetic accessibility

TIER-1 — 8 candidates
Expert curation + multi-parameter optimization. Proteins are dynamic: static docking can't confirm stability. Each candidate simulated 20 ns x 3 replicates at 310 K, 0.15 M NaCl (physiological)

Molecular Dynamics — 5 stable binders
520 ns total simulation, OpenMM GPU-accelerated. Gold standard for binding free energy: quantifies how R702W mutation alters each compound's affinity using 120 lambda windows + MBAR estimator

Free Energy Perturbation — 2 FEP-validated
Alchemical ΔΔG calculation across WT and R702W systems. Reveals opposite mutation sensitivity: bufadienolide binds tighter to WT, pristimerin binds tighter to R702W, enabling mutation-specific therapy

Opposite mutation sensitivity profiles
2 LEADS

99.98% reduction: 9,566 → 2 validated leads

Figure 10: Screening cascade showing 9,566 compounds filtered to 2 validated leads. Graphic created by finalist using Canva, 2026.

1. PREPARATION
Structure prediction and compound library assembly

No experimental NOD2 structure exists → used **AlphaFold2** (CASP14 winner).
Predicted full-length **NOD2** (**1,040 residues**) from UniProt **Q9HC29**.
Focus: **LRR domain** (residues 744-1040), **pLDDT >90** = high confidence.
Prepared with polar hydrogens + Gasteiger-Marsili partial charges.

e-Drug3D: **2,152 FDA-approved drugs** (known safety profiles).
COCONUT: **7,414 natural products** (unique scaffolds).
3D structures via **RDKit ETKDG** + **MMFF94** minimization.

Total: **9,566 unique compounds** for screening.

2. GNINA
Fast docking to rank binders

GNINA: next-gen docking with **CNN scoring** trained on **PDBbind**.
Unlike AutoDock Vina/Glide, CNNs learn binding patterns from real structures.

Grid: **30×30×30 Å** centered on LRR domain (residues 750-1040).
Exhaustiveness = 32 (4× default) for thorough conformational sampling.
9 binding poses per ligand to capture different orientations.

CNN affinity scores: **2.47–7.34 kcal/mol** (mean: **4.57**).

All **9,566 compounds** docked and ranked.

3. NOD2-SCOUT
ML reranks hits (docking alone has ~0.5 correlation with experiment)

XGBoost classifier (500 trees, depth=6, lr=0.05) on docking pseudo-labels:
Top 20% = binders, bottom 20% = non-binders, stratified by **5 MW bins**.
856 labeled compounds (728 training, 128 test).

Features: **2,048-bit Morgan fingerprints** + **10 descriptors** = **2,058 total**.

5-fold scaffold-split CV: entire scaffolds held out (stricter than random).

Performance: **AUC = 0.85–0.93**. Shuffled control: **AUC = 0.50** (random).

Filtered to top **2,129 compounds**.

4. ADMET
Safety and drug-likeness filter

A strong binder is useless if it can't reach the target or causes toxicity.

Lipinski's Rule of 5: MW **≤500**, LogP **≤5**, HBD **≤5**, HBA **≤10**.

Veber's rules: **TPSA ≤140 Ų** (oral bioavailability), **rotatable bonds ≤10**.

PAINS: removes false-positive-prone compounds (aggregation/redox).

Brenk filters: flags known toxic substructures (>2 alerts = removed).

All filtering via **RDKit** cheminformatics toolkit.

Result: **148 of 150** passed (**98.7%**).

5. TIER-1
Expert curation (computation can't capture everything)

Manual curation of **148 ADMET-passing compounds** using multi-parameter optimization.

Structural diversity: Tanimoto clustering on Morgan fingerprints.
Diverse scaffolds = higher odds of success in experimental validation.

Binding pose quality: H-bonds to **GLU1008**, **ASP1011**, **ARG1037**;
π-stacking; hydrophobic burial. Strained/exposed poses deprioritized.

Synthetic accessibility: FDA drugs available; natural products vary.

Selected **8 candidates** → **5** advanced to MD.

6. MOLECULAR DYNAMICS
Checks if binding is stable over time (proteins are dynamic!)

OpenMM GPU-accelerated (~100 ns/day on RTX). **520 ns** total simulation.
Force fields: **AMBER14** (protein), **OpenFF 2.1.0** (ligand), **TIP3P-FB** water.
20 ns/system at 310 K, **3 replicates** each. 0.15 M NaCl (physiological).

Pocket occupancy: % frames ligand stays within **5 Å** of start position.

Negative control **Cysteamine** (rank #2056/2129): **0% occupancy**.
Validates pocket is specific—high occupancy = genuine binding.

Top **2 stable** (**70-80% occupancy**, ~2 H-bonds).

7. FREE ENERGY PERTURBATION
Gold standard for quantifying binding affinity differences

FEP "alchemically" transforms ligand → nothing, measuring ΔG in bound vs free state.
120 lambda windows total (20/leg × 4 legs × 2 compounds). **MBAR** estimator.
ΔΔG = ΔG(R702W) − ΔG(WT): positive = mutation weakens binding.

Febuxostat: ΔG(WT)=**−10.36**, ΔG(R702W)=**−8.02** kcal/mol.
ΔΔG = **+2.34 kcal/mol** (**8σ**, p<0.001) → **50× weaker** to mutant.

Bufadienolide: ΔG(WT)=**−15.22**, ΔG(R702W)=**−15.66** kcal/mol.
ΔΔG = **−0.44 kcal/mol** (**1.8σ**, NS) → binding **unaffected**.

Key insight: mutation effects are **ligand-dependent**!

8. 2 VALIDATED LEADS
Two compounds with opposite mutation sensitivity profiles

99.98% reduction: **9,566** → **2** FEP-validated leads.

LEAD 1: Febuxostat (FDA-approved gout drug)
Mutation-SENSITIVE: 50× weaker to R702W. Mutation is **79.4 Å away**!
→ Probe for studying allosteric effects of distant mutations.

LEAD 2: Bufadienolide (CID 10120, natural cardiac glycoside)
Mutation-INSENSITIVE: binds equally to WT and R702W.
→ Starting scaffold for mutation-resistant therapeutics.

Mutation effects are ligand-specific → precision medicine insight.

RESULTS

MACHINE LEARNING VALIDATION
Scaffold-split cross-validation confirms model learned real patterns, not random noise

What This Proves

AUC = 0.85–0.93 vs shuffled **0.50**

856 compounds labeled | **728** train / **128** test | **2,058** features

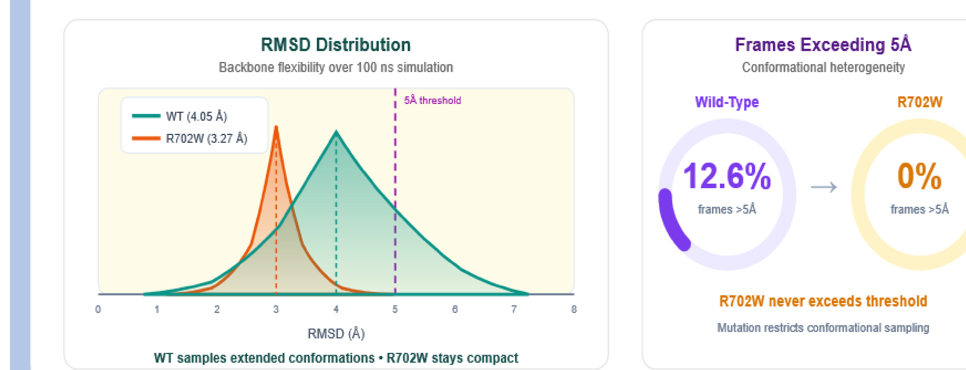
Scaffold-split CV keeps similar molecules together during testing
Model learned **generalizable chemistry patterns**, not memorizing scaffolds

Shuffled control (AUC = 0.50) confirms no data leakage or overfitting

Figure 11: ROC curve and 5-fold cross-validation AUC scores. Graphs created by finalist using Microsoft Excel, 2026.

RESULTS COUNT.

APO DYNAMICS: WT vs R702W
The R702W mutation rigidifies the protein, reducing conformational sampling



Mechanistic Insight

R702W = MORE RIGID than wild-type

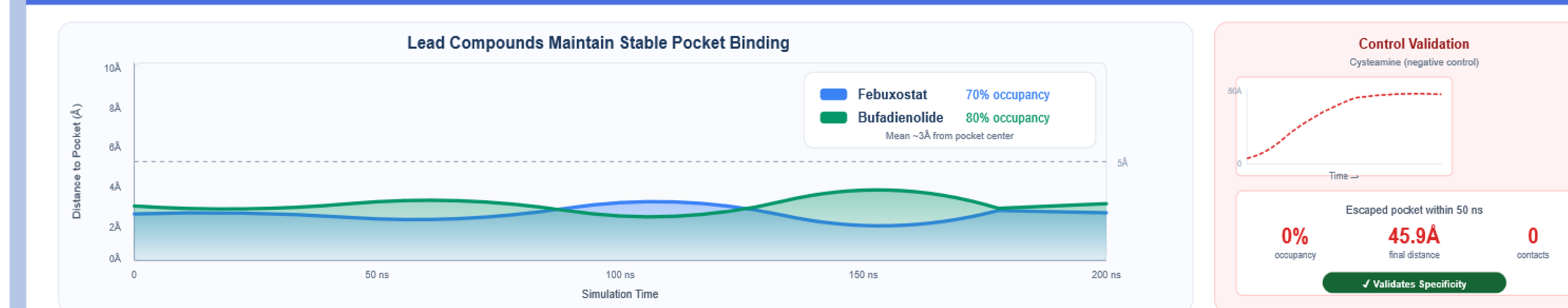
Total Simulation	Replicates	Per System
100 ns	5 (2+3)	20 ns

Domain Flexibility (RMSF):
HD2: **1.46 → 1.23 Å** LRR: **1.36 → 1.12 Å**

Explains how mutation 79.4 Å away affects ligand binding allosterically

Figure 12: RMSD distribution, frames exceeding 5Å, and mechanistic insight for WT vs R702W molecular dynamics. Graphs created by finalist using Microsoft Excel, 2026.

MOLECULAR DYNAMICS VALIDATION
520 ns total simulation across lead compounds and negative control

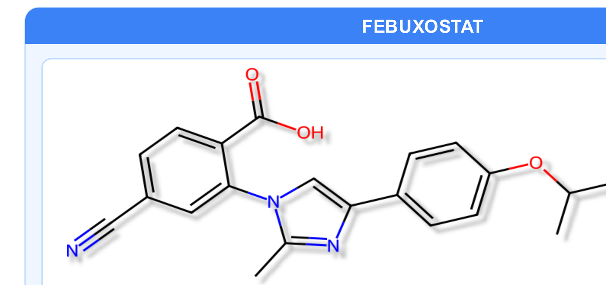


Febuxostat (FDA-approved) and **Bufadienolide** (natural product) both maintain stable binding throughout 200 ns simulations, with **70-80% pocket occupancy**, ~2 hydrogen bonds per frame, and 6-7 stable residue contacts (>50% occupancy).
The negative control **Cysteamine** (bottom 3% CNN score) escaped to 45.9Å within 50 ns, **validating assay specificity**.

Figure 13: Lead compound pocket binding stability over 200 ns simulation. Graph created by finalist using Canva, 2026.

LEAD COMPOUNDS
Two chemically diverse hits from FDA drug library and natural products database

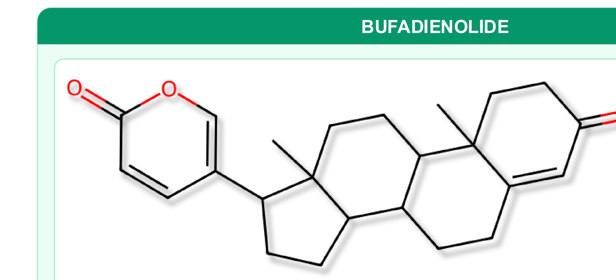
FEBUXOSTAT — FDA Drug



Source: e-Drug3D
MW: 315.37
LogP: 2.39
TPSA: 86 Å²
CNN: 5.72
Tier 1, #1
✓ Lipinski
70% Occupancy

Figure 14: Febuxostat 2D molecular structure. Image created by finalist using MolView, 2026.

BUFADIENOLIDE — Natural Product

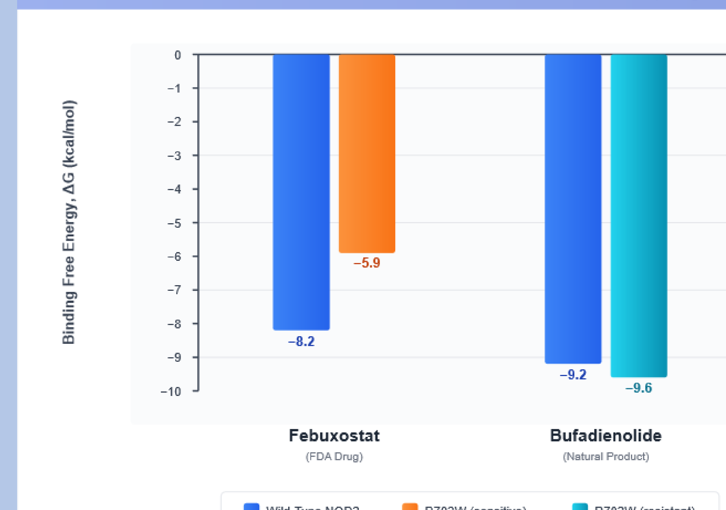


Source: COCONUT
MW: 402.53
LogP: 3.36
TPSA: 91 Å²
CNN: 6.76
Tier 2, #19
✓ Lipinski
80% Occupancy

Figure 15: Bufadienolide 2D molecular structure. Image created by finalist using MolView, 2026.

Febuxostat, an FDA-approved gout medication, represents a drug repurposing opportunity that is already proven safe in humans.
Bufadienolide, a natural cardiac glycoside, offers a novel scaffold for NOD2 modulation. Both passed Lipinski's Rule of Five with stable MD binding.

FREE ENERGY PERTURBATION RESULTS
Comparing Drug Binding Affinity: Wild-Type vs. R702W Mutant NOD2



BIOLOGICAL INTERPRETATION

FEBUXOSTAT — Mutation-Sensitive
ΔΔG = +2.34 kcal/mol | WT: −8.2 → R702W: −5.9 kcal/mol
R702W mutation disrupts binding pocket → 50× affinity loss
→ Reduced efficacy in Crohn's patients with NOD2 variants

BUFADIENOLIDE — Mutation-Resistant
ΔΔG = −0.44 kcal/mol | WT: −9.2 → R702W: −9.6 kcal/mol
Binding affinity preserved — no significant change with mutation
→ Ideal candidate for precision medicine in NOD2+ patients

FEP Protocol
100 ns simulations • 12 λ-windows • OPLS4 force field • Error = SEM (n=3)

Figure 16: FEP binding free energy comparison for WT vs R702W NOD2. Graph created by finalist using Microsoft Excel, 2026..

DISCUSSION

Why does Febuxostat fail at R702W?

R702W substitutes charged arginine (+1) with neutral tryptophan, eliminating 2-3 hydrogen bonds. The lost positive charge disrupts electrostatic networks propagating **79.4 Å** to the LRR binding pocket. FEP: **ΔΔG = +2.34 kcal/mol** (p<0.001) = **50× weaker binding** to the R702W mutant protein compared to wild-type NOD2.

Allosteric mechanism: rigidification

APO molecular dynamics simulations show R702W **rigidifies** the NOD2 protein: 0% vs 12.6% of trajectory frames exceed 5Å RMSD. Mean RMSD values: 3.27 Å vs 4.05 Å (wild-type is more flexible). Reduced conformational flexibility impairs **induced-fit binding** for ligands requiring protein adaptation to bind optimally. Challenges prior literature that described R702W as "destabilizing" the protein.

Limitations

Computational study requires **SPR binding** and **NF-κB reporter assay** validation. Only R702W tested; G908R and L1007fs remain. No crystal structure available.

Why does Bufadienolide maintain binding?

Completely distinct contact profile from Febuxostat: **ARG1034: 82%** vs 3% occupancy and **ARG1037: 81%** vs 71%. The hydrophobic steroid scaffold relies primarily on π-stacking interactions rather than mutation-sensitive electrostatic networks. FEP result: **ΔΔG = −0.44 kcal/mol** = **binding preserved** in mutant.

Biological significance

Bufadienolide's mutation-resistant binding suggests potential **restoration of NOD2 signaling** in R702W carriers, a functional rescue for impaired innate immunity. Could enable **proper bacterial clearance** in the 15% of Crohn's patients carrying R702W who currently lack targeted therapies.

Clinical implications

Mutation effects on drug binding are **ligand-dependent**, not universal across all compounds. Screening against wild-type protein alone would fail 15% of patients.

CONCLUSIONS

Hypothesis supported

Deep learning combined with molecular dynamics successfully identified mutation-resistant NOD2 binders. FEP quantified differential binding affinity between wild-type and R702W mutant proteins with high statistical significance (p<0.001, 8σ effect).

Key findings

1. Screened 9,566 compounds → identified 2 validated lead binders for NOD2 protein
2. Febuxostat (FDA-approved for gout): binds NOD2 but **50× weaker** in R702W mutant
3. Bufadienolide (natural product): **mutation-resistant** binder (ΔΔG = −0.44 kcal/mol)
4. R702W causes allosteric rigidification propagating 79.4 Å from mutation to pocket
5. Cysteamine negative control (0% occupancy) validates computational assay specificity

Significance

First computational drug screen targeting the NOD2 LRR domain in scientific literature.
First FEP study quantifying how Crohn's-associated mutations affect NOD2 binding.
Febuxostat identified as FDA-approved drug repurposing candidate for Crohn's disease.

Future directions

- SPR/ITC binding assays to experimentally confirm computational binding affinities
- NF-κB reporter functional assays in NOD2-expressing cell lines to test activation
- Extend FEP analysis to G908R mutation to cover ~50% of NOD2-Crohn's patients
- Screen bufadienolide structural analogs for improved drug-likeness and lower toxicity
- X-ray crystallography to validate computationally predicted binding poses and contacts

References

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Acknowledgements: NOD2 structure from AlphaFold DB (AF-Q9HC29). Compound libraries from COCONUT & e-Drug3D databases. R702W mutation introduced computationally. All molecular visualizations created by finalist using PyMOL and Python.